Density Functional Perturbation Theory and applications: Raman spectroscopy of Ar impurities in Corundum Al_2O_3

Presented by Alessandro Ciavatta,
with the supervision of Lorenzo Paulatto

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Campus Pierre and Marie Curie
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Introduction

Density Functional Theory (DFT) allows the calculation of the ground state energy and density of a many-electron system with an accuracy determined by the choice of the approximation for the exchange-correlation (xc) effects. The electron dispersion is obtained directly by applying the Kohn and Sham procedure, while phonons can be computed by diagonalizing the matrix of the energy derivative with respect to each displacement. By examining energies near the equilibrium position, one can construct this matrix using finite differences. An alternative, powerful, and general approach is the application of quantum perturbation theory within the DFT framework. This approach shares the same scaling as a single ground state calculation and can handle perturbations in extended systems with incommensurable wave vectors. This work aims to demonstrate this approach and its applications in calculating the following properties: phonon spectrum, Raman cross-section, and phonon lifetimes. We first describe the mechanism, along with some equivalent variations. Then, we discuss the power and predictability of simulated vibrational spectra in Corundum with and without Ar impurities.

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1 Density Functional Theory

1.1 Adiabatic approximation

The full, non-relativistic hamiltonian of a system composed of nuclei and electrons $H = T_e + V_e + V_{ext} + T_N + V_N$ has the following contributions:

- electrons kinetic energy $T_e = -\frac{1}{2}\sum_i \nabla_i^2$
- electron-electron interaction $V_e = \frac{1}{2}\sum_{ij} |\boldsymbol{r}_i - \boldsymbol{r}_j|^{-1}$
- nucleus-electron interaction $V_{ext} = -\sum_{iI} Z_I |\boldsymbol{r}_i - \boldsymbol{R}_I|^{-1}$
- nuclei kinetic energy $T_N = -\frac{1}{2}\sum_I \frac{\nabla_I^2}{M_I}$
- nucleus-nucleus interaction $V_N = \frac{1}{2}\sum_{IJ} Z_I Z_J |\boldsymbol{R}_I - \boldsymbol{R}_J|^{-1}$

Atomic units are used throughout the entire dissertation. The most common approximation, known as the *adiabatic* or *Born-Oppenheimer* approximation, involves first solving the Hamiltonian that includes the degrees of freedom of the electrons, while treating the positions of the ions, denoted as $\boldsymbol{R}_I$, as fixed parameters. This approximation is always justified since the kinetic term of the nuclei can be treated as a perturbation ($1/M_I \ll 1$). The potential V_N does not depend on the electronic degrees of freedom and can be determined simply by considering the positions of the nuclei. Its effect is to shift the energies of the system, with the positions of the ions kept frozen. The following discussion focuses on this system, which can be described by the Hamiltonian $H = T_e + V_e + V_{ext}$.

1.2 Hohemberg-Kohn Theorems

The full many-body Schrödinger equation for a system of N electrons is solved by determining the many-body eigenfunctions $\Psi(\boldsymbol{r}_i)$, which involve all $3N$ electronic degrees of freedom. Due to the Coulomb repulsion between electrons, the Hamiltonian cannot be separated into single particle terms, and the wavefunction Ψ cannot be factorized. Hohenberg and Kohn [1] demonstrated that all the relevant properties of a system, including its excited state electronic properties, can be derived from the ground state electron density $n_0(\boldsymbol{r})$. Hohenberg and Kohn formulated two theorems that establish the effectiveness of this approach for addressing the many-body electron problem:

- **Theorem I:** The external potential $V_{ext}(\boldsymbol{r})$ of a system of interacting electrons is uniquely determined, up to a constant, by the ground state

electron density $n_0(\boldsymbol{r})$. The electronic hamiltonian $H = T_e + V_e + V_{ext}$ is fully determined by the external potential since the forms of the kinetic and Coulomb repulsion terms are independent from the system's structure. By constructing the corresponding Hamiltonian from $n_0(\boldsymbol{r})$, one can access the ground and excited states and energies by solving the many-body Schrödinger equation.

Proof: Supposing that there are two systems with external potentials V_{ext} and V'_{ext}, associated with the same ground state density $n_0(\boldsymbol{r})$. The two systems are then characterized by different Hamiltonians H and H', with different ground states Ψ and Ψ'. Thus,

$$E = \langle \Psi | H | \Psi \rangle < E' + \langle \Psi' | H - H' | \Psi' \rangle$$
$$E' = \langle \Psi' | H' | \Psi' \rangle < E - \langle \Psi | H - H' | \Psi \rangle .$$

The inequality is strict only in the non-degenerate case, but the demonstration can be extended to encompass these situations as well. The difference between the Hamiltonians boils down to the discrepancy in external potentials, which yields the same outcome when evaluated for both Ψ and Ψ':

$$\langle \Psi | H - H' | \Psi \rangle = \langle \Psi' | H - H' | \Psi' \rangle = \int d^3r \, [V_{ext}(\boldsymbol{r}) - V'_{ext}(\boldsymbol{r})] \, n_0(\boldsymbol{r}).$$

These three constraints cannot be fulfilled simultaneously; hence, the hypothesis is incorrect.

- **Theorem II:** $T_e + V_e$ can be explicitly constructed as functionals of the electron density $n(\boldsymbol{r})$. The ground state electron density $n_0(\boldsymbol{r})$ minimizes the functional.

$$E[n] = T_e[n] + V_e[n] + \int d^3r \, V_{ext}(\boldsymbol{r}) n(\boldsymbol{r}) \tag{1}$$

and the minimum value corresponds to the ground state energy.

Proof: The construction of the energy functional is derived from Levy and Lieb [2, 3]. The ground state energy is obtained by minimizing the energy with respect to every possible wavefunction Ψ. Alternatively, one can first minimize within the restricted space $A(n)$, which consists of all wavefunctions with the same density $n(\boldsymbol{r})$, and then minimize with respect to the densities. As a result, the universal functional arises from the initial minimization process: $T_e[n] + V_e[n] = \min_{A(n)} \langle \Psi | T_e + V_e | \Psi \rangle$. Each density can then be associated with the wavefunction Ψ that minimizes the universal functional. In particular,

$$E_0 = E[n_0] = \langle \Psi_0 | H | \Psi_0 \rangle < \langle \Psi | H | \Psi \rangle = E[n]$$

1.3 Kohn-Sham system

Let's consider a system of independent particles, in which the Hamiltonian can be separated and the trial wavefunction is the product of single orbitals ψ_i. The density and the kinetic term are expressed as:

$$n(\boldsymbol{r}) = \sum_i |\psi_i(\boldsymbol{r})|^2 \tag{2}$$

$$T_s = -\frac{1}{2}\sum_i \langle\psi_i|\nabla^2|\psi_i\rangle\,. \tag{3}$$

The most natural way to treat charged particles as if they were independent is the *Hartree* approximation: each particle is subject to a Coulomb potential generated by all the other particles. The contribution to the total energy is given by the integral

$$E_H[n] = \frac{1}{2}\int d^3r d^3r'\,\frac{n(\boldsymbol{r})n(\boldsymbol{r'})}{|\boldsymbol{r}-\boldsymbol{r'}|}, \tag{4}$$

which adds adds a spurious interaction between the single particle and itself. While this term is an explicit functional of the density, the kinetic term is guaranteed to have the same functional dependence, simply following Levy and Lieb's construction. This approach is missing two important ingredients:

- *Exchange:* The ground state many-body fermionic wavefunction is expected to be antisymmetric under the exchange of particles. One can remedy this inadequacy by explicitly antisymmetrizing the trial wavefunction, which is expected to yield lower energy expectation values and be qualitatively more similar to the true ground state function.

- *Correlation:* Within this picture, the only electronic configuration taken into account is that in which the lowest energy orbitals are occupied. By allowing the electrons to interact, we expect to see excited electronic configurations with a non-negligible weight in the trial wavefunction. Relaxing this constraint is expected to further lower the energy expectation value.

The density functional approach groups these many-body effects into a single exchange-correlation energy:

$$E_{xc}[n] = T_e[n] - T_s[n] + V_e[n] - E_H[n]. \tag{5}$$

It's important to notice that every single piece is a functional of the density. The true energy functional in Eq. (1) can be rewritten as

$$E[n] = T_s[n] + E_H[n] + \int d^3r\, V_{ext}(\boldsymbol{r})n(\boldsymbol{r}) + E_{xc}[n]. \tag{6}$$

The central idea of Kohn and Sham [4] is based on the following assumption: a system of interacting electrons described by an external potential can be mapped to an equivalent system of non-interacting particles with a different effective potential that shares the same ground state density. The effective potential results from the minimization of the energy with respect to the orbitals:

$$V_{SCF}(\boldsymbol{r}) = V_{ext}(\boldsymbol{r}) + \int d^3r' \, \frac{n(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} + \frac{\delta E_{xc}}{\delta n(\boldsymbol{r})} \tag{7}$$

$$H_{SCF}(\boldsymbol{r}) = -\frac{1}{2}\nabla^2 + V_{SCF}(\boldsymbol{r}); \quad H_{SCF}\psi_i(\boldsymbol{r}) = \epsilon_i \psi_i(\boldsymbol{r}), \tag{8}$$

with the last term of the potential also called $v_{xc}(\boldsymbol{r})$. These equations are solved self-consistently, starting with a guess for the density, calculating the effective potential, solving for the orbitals and evaluating the density through Eq. (2). The energy functional should give lower and lower results at each cycle, eventually stopping to improve substantially once convergence is achieved. Although the functional dependence of the exchange-correlation is still unknown, some approximations have proven to be successful. The simplest one is the local density approximation: at each point $\boldsymbol{r}$ the exchange-correlation energy per particle is approximated with the homogeneous electron gas one with density $n(\boldsymbol{r})$. While the exchange functional is known analytically, correlation is fitted with Quantum Monte Carlo data. A generally more predictive approximation is the Generalized Gradient Approximation (GGA), by which the functional derivative is defined in a neighborhood of $\boldsymbol{r}$.

1.4 Density Functional Perturbation Theory (DFPT)

The electronic ground state density [5], [6] is the key ingredient for lattice dynamics. The lattice Hamiltonian, within the adiabatic approximation, is written in terms of $n_0(\boldsymbol{r})$ in the actual lattice configuration $\boldsymbol{R}$:

$$H = T_N + V_N(\{\boldsymbol{R}\}) + E([n_0], \{\boldsymbol{R}\}). \tag{9}$$

The nuclei move under the action of forces, which are simply given by the gradient of the energy expectation value. A huge simplification of the problem of the lattice dynamics comes from the *Hellmann-Feynman theorem* or force theorem, which states that the knowledge of the variational ground state at position $\{\boldsymbol{R}\}$ is sufficient to calculate the gradients:

$$\boldsymbol{F}_I = -\frac{\partial E}{\partial \boldsymbol{R}_I} = -\frac{\partial \langle \Psi_{\{\boldsymbol{R}\}} | H | \Psi_{\{\boldsymbol{R}\}} \rangle}{\partial \boldsymbol{R}_I} = -\langle \Psi_{\{\boldsymbol{R}\}} | \frac{\partial H}{\partial \boldsymbol{R}_I} | \Psi_{\{\boldsymbol{R}\}} \rangle . \tag{10}$$

The same theorem can be written in functional form and applied to the DFT ground state: the contribution coming from the differentiation of the electronic energy with respect to the density is zero at the ground state density, hence only the explicit derivative remains:

$$\boldsymbol{F}_I = -\frac{\partial V_N}{\partial \boldsymbol{R}_I} - \frac{\partial E}{\partial \boldsymbol{R}_I} = -\frac{\partial V_N}{\partial \boldsymbol{R}_I} - \int d^3r \, \frac{\partial V_{ext}(\{\boldsymbol{R}\}, r)}{\partial \boldsymbol{R}_I} n(\boldsymbol{r}). \tag{11}$$

By following the forces, one can fully reproduce the system's dynamics or relax it to the equilibrium configuration. Near the equilibrium, in the harmonic approximation, the Hamiltonian reduces to the sum of independent harmonic oscillators, called *normal modes*. The frequency and polarization of these oscillations are obtained by diagonalizing the interatomic force constants (IFC) matrix:

$$\frac{\partial^2 E}{\partial \boldsymbol{R}_I \partial \boldsymbol{R}_J} = \frac{\partial^2 V_N}{\partial \boldsymbol{R}_I \partial \boldsymbol{R}_J} + \int d^3r \, \left[\delta_{IJ} \frac{\partial^2 V_{ext}}{\partial \boldsymbol{R}_I \partial \boldsymbol{R}_J} n(\boldsymbol{r}) + \frac{\partial V_{ext}}{\partial \boldsymbol{R}_I} \frac{\partial n(\boldsymbol{r})}{\partial \boldsymbol{R}_J} \right]. \tag{12}$$

As a consequence of Eq. (11), the IFCs depend on the response of the ground state electronic density to the atomic displacement only up to *linear* order, which dramatically simplifies the problem. Within perturbation theory, first-order variations are obtained by linearizing the Kohn and Sham's equations in Eqs. (2), (7)–(8):

$$\Delta n(\boldsymbol{r}) = 4\mathrm{Re} \sum_{n=1}^{N/2} \psi_n^*(\boldsymbol{r}) \Delta\psi_n(\boldsymbol{r}) \tag{13}$$

$$\Delta V_{SCF}(\boldsymbol{r}) = \Delta V_{ext}(\boldsymbol{r}) + \int d^3r' \frac{\Delta n(\boldsymbol{r'})}{|\boldsymbol{r} - \boldsymbol{r'}|} + \frac{dv_{xc}}{dn} \Delta n(\boldsymbol{r}) \tag{14}$$

$$(H_{SCF} - \epsilon_n) \left|\Delta\psi_n\right\rangle = -(\Delta V_{SCF} - \Delta\epsilon_n) \left|\psi_n\right\rangle, \tag{15}$$

where Δ stands for a variation with respect to any set of parameters. One can use the standard perturbation theory formula for $\Delta\psi_n(\boldsymbol{r})$ to write the density as:

$$\Delta n(\boldsymbol{r}) = 4\mathrm{Re} \sum_{n=1}^{N/2} \sum_{m \neq n} \psi_n^*(\boldsymbol{r}) \psi_m(\boldsymbol{r}) \frac{\langle \psi_m | \Delta V_{SCF} | \psi_n \rangle}{\epsilon_n - \epsilon_m}. \tag{16}$$

It is easy to see that when the index m runs over the valence band, we have two contributions that always cancel each other. Since the projector over the conduction band P_c commutes with the Hamiltonian, and $P_c \left|\Delta\psi_n\right\rangle = \left|\Delta\psi_n\right\rangle$, projecting both sides of Eq. (15) we obtain:

$$(H_{SCF} - \epsilon_n) \left|\Delta\psi_n\right\rangle = -P_c \Delta V_{SCF} \left|\psi_n\right\rangle \tag{17}$$

Expressions of this kind are called *Sternheimer equations*. The operator $(H_{SCF} - \epsilon_n)$ cannot be directly inverted for a direct solution of the equation through Green's function technique, as it becomes singular when evaluated at $|\psi_n\rangle$. To avoid this singularity, the projection is usually introduced in equations as $P_c(H_{SCF} - \epsilon_n)P_c$. The main disadvantage in Eq. (16) is that we need to solve the unperturbed Hamiltonian for an arbitrary amount of unperturbed empty orbitals to evaluate the perturbed density. On the other hand, the Sternheimer equation can be directly solved on occupied states: $P_c = 1 - P_v = 1 - \sum_v |\psi_v\rangle \langle\psi_v|$, where v stands for valence states.

1.5 Extended systems

Extended systems are described by a unit cell ($\boldsymbol{a}_1$, $\boldsymbol{a}_2$, $\boldsymbol{a}_3$). The Hamiltonian is invariant under translation of $\boldsymbol{R_l} = l_1\boldsymbol{a}_1 + l_2\boldsymbol{a}_2 + l_3\boldsymbol{a}_3$, where $\boldsymbol{l}$ is a set of integers. Inside the unit cell, atomic positions are described by the basis $\boldsymbol{\tau}_s$. Consequently, in the basis diagonal to both the Hamiltonian and the translation operator, electronic states are classified by the wave-vector $\boldsymbol{k}$ and the band index ν. $\boldsymbol{k}$ belongs to the reciprocal lattice ($\boldsymbol{b}_1$, $\boldsymbol{b}_2$, $\boldsymbol{b}_3$), defined by the orthogonality condition $\boldsymbol{a}_i \cdot \boldsymbol{b}_j = \delta_{ij}$. It labels solution of the translation operator eigenvalue equation; ν is an integer which labels different solutions of the stationary Schrödinger equation. These states can be written in the so-called Bloch form as $\psi_{\boldsymbol{k}\nu}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}} u_{\boldsymbol{k}\nu}(\boldsymbol{r})$, where u is a lattice-periodic function. In the harmonic approximation, the Hamiltonian in Eq. (9) is expanded up to second-order as:

$$H = T_N + E_0 + \frac{1}{2}\sum_{s\boldsymbol{l},t\boldsymbol{l'}} \frac{\partial^2 E}{\partial \boldsymbol{u}_s(\boldsymbol{l})\partial \boldsymbol{u}_t(\boldsymbol{l'})}\bigg|_{u=0} \boldsymbol{u}_s(\boldsymbol{l})\boldsymbol{u}_t(\boldsymbol{l'}), \tag{18}$$

where $\boldsymbol{u}_s(\boldsymbol{l})$ is the displacement of the atom s in the cell $\boldsymbol{l}$. This Hamiltonian breaks the periodiciy of the lattice and couples different ionic displacements. Although, due to periodicity at $u = 0$, the real-space IFCs depend only on the difference $\boldsymbol{l} - \boldsymbol{l'}$:

$$C_{st}(\boldsymbol{l} - \boldsymbol{l'}) = \frac{\partial^2 E}{\partial \boldsymbol{u}_s(\boldsymbol{l})\partial \boldsymbol{u}_t(\boldsymbol{l'})}\bigg|_{u=0}. \tag{19}$$

The Hamiltonian can be decoupled in reciprocal space, by defining a monochromatic displacement with wavevector $\boldsymbol{q}$ as $\boldsymbol{u}_s(\boldsymbol{l}) = \boldsymbol{u}_s(\boldsymbol{q})e^{i\boldsymbol{q}\cdot\boldsymbol{R_l}}$ and diagonalizing the Fourier transformed IFCs:

$$C^{\alpha\beta}_{st}(\boldsymbol{q}) = \sum_{\boldsymbol{l}} e^{-i\boldsymbol{q}\cdot\boldsymbol{R_l}} C^{\alpha\beta}_{st}(\boldsymbol{l}). \tag{20}$$

Each normal mode is labeled by a specific wavevector $\boldsymbol{q}$ and the branch index ν, which counts the number of modes for a specific $\boldsymbol{q}$. The advantage of working with DFPT is that the perturbation and the response share the respective wavelength, thanks to the linearity of the Eqs. (13)–(15). To solve the phonon equation for a specific wavevector $\boldsymbol{q}$, one can calculate the derivative of the potential as follows:

$$\frac{\partial V_{ext}(\boldsymbol{r})}{\partial \boldsymbol{u}_s(\boldsymbol{q})} = -\sum_{\boldsymbol{l}} \frac{\partial V_{ext}(\boldsymbol{r} - \boldsymbol{R_l} - \boldsymbol{\tau}_s)}{\partial \boldsymbol{r}} e^{i\boldsymbol{q}\cdot\boldsymbol{R_l}}. \tag{21}$$

The resulting variation is in the Bloch form: $\Delta V^{\boldsymbol{q}}_{ext}(\boldsymbol{r} + \boldsymbol{R}) = e^{i\boldsymbol{q}\cdot\boldsymbol{R}} \Delta V^{\boldsymbol{q}}_{ext}(\boldsymbol{r})$. The perturbation carries a momentum $\boldsymbol{q}$, since it shares the orthogonality property of the electronic Bloch functions:

$$\langle \psi_{n'\boldsymbol{k'}} | \Delta V^{\boldsymbol{q}}_{ext} | \psi_{n\boldsymbol{k}} \rangle = \sum_{\boldsymbol{l}} e^{i(\boldsymbol{k}+\boldsymbol{q}-\boldsymbol{k'})\cdot\boldsymbol{R_l}} f(\boldsymbol{k}, \boldsymbol{k'}, n, n', \boldsymbol{q}) \tag{22}$$

$$= N\delta_{\boldsymbol{k}+\boldsymbol{q},\boldsymbol{k'}} f(\boldsymbol{k}, n, n', \boldsymbol{q}), \tag{23}$$

Where N represents the number of cells and f denotes the integral of the periodic parts of the three Bloch functions. As a result, the perturbed eigenfunctions have defined momentum $\boldsymbol{k} + \boldsymbol{q}$:

$$|\Delta\psi_{n\boldsymbol{k}}\rangle = \sum_{n'} |\psi_{n'\boldsymbol{k}+\boldsymbol{q}}\rangle \frac{\langle \psi_{n'\boldsymbol{k}+\boldsymbol{q}} | \Delta V^{\boldsymbol{q}}_{ext} | \psi_{n\boldsymbol{k}} \rangle}{\epsilon_{n\boldsymbol{k}} - \epsilon_{n'\boldsymbol{k}+\boldsymbol{q}}} \tag{24}$$

and the density in Eq. (13) is also expressed in Bloch form as $\Delta n_{\boldsymbol{q}}(\boldsymbol{r}) = e^{i\boldsymbol{q}\cdot\boldsymbol{r}} \Delta\tilde{n}_{\boldsymbol{q}}(\boldsymbol{r})$, where $\tilde{n}$ represents a lattice periodic function. Similarly, the entire SCF potential in Eq. (14) takes on the same Bloch form. The Sternheimer equation in Eq. (17) is solved by expanding in plane waves of reciprocal lattice vectors, just as in the unperturbed case.

1.6 Homogeneous electric fields

In metals, valence electrons are free to move: under the action of an electric field $\boldsymbol{\mathcal{E}}$, electrons start to flow under the action of the potential $V = -e\boldsymbol{\mathcal{E}} \cdot \boldsymbol{r}$ until the electric field inside the material vanishes. On the other hand, insulators respond to a small homogeneous electric field by developing a polarization $\boldsymbol{\mathcal{P}}$, which only partially screens the electric field. The polarization in insulators arises from two main contributions. The first contribution originates from the redistribution of electron density and is governed by the electronic susceptibility $\chi^{\infty}_{\alpha\beta} = \partial\mathcal{P}_\alpha / \partial\mathcal{E}_\beta$, which is calculated with the ions held fixed. The second contribution arises from the ionic displacements that induce a

macroscopic polarization. This phenomenon occurs in insulators exhibiting polar behavior, where the displacement of an ion generates an electric dipole within the unit cell. Furthermore, dipoles generated by longitudinal displacements (LO phonons) are unable to cancel out in the long-wavelength limit and give rise to an additional contribution to the polarization, governed by the response $Z_s^{*\alpha\beta} = \partial \mathcal{P}_\alpha / \partial u_s^\beta(\boldsymbol{q} = 0)$, calculated at vanishing external electric fields. This responses are also known as *Born effective charges*. We first analyse the electronic contribution to the polarization:

$$\boldsymbol{\mathcal{P}} = -\frac{1}{V} \int d^3r \, \Delta n(\boldsymbol{r}) \boldsymbol{r}, \tag{25}$$

where the variation of the density, from now on, is intended to come from the variation of the electric field $\boldsymbol{\mathcal{E}}$. This expression becomes ill-defined in periodic systems, as the choice of the origin can affect the polarization. However, in the linear regime, we can accurately define the position operator $\boldsymbol{r}$ in terms of lattice-periodic operators. This can be done using the commutation relation:

$$\langle \psi_m | \boldsymbol{r} | \psi_n \rangle = \frac{\langle \psi_m | [H_{SCF}, \boldsymbol{r}] | \psi_n \rangle}{\epsilon_m - \epsilon_n} = -\frac{\hbar^2}{m} \frac{\langle \psi_m | \partial / \partial \boldsymbol{r} | \psi_n \rangle}{\epsilon_m - \epsilon_n}, \tag{26}$$

where ϵ_m and ϵ_n are the energies of the respective states. Since we are only interested in off-diagonal elements to calculate linear responses, the denominator is always nonzero. Eq. (25), written in KS orbitals through Eq. (13), becomes:

$$\mathcal{P}_\alpha = -\frac{4}{V} \sum_n \langle \psi_n | r_\alpha | \Delta \psi_n \rangle \tag{27}$$

$$= -\frac{4}{V} \sum_n \left[\sum_{m \neq n} \frac{\langle \psi_m | [H_{SCF}, r_\alpha] | \psi_n \rangle}{\epsilon_m - \epsilon_n} \langle \psi_m | \right] | \Delta \psi_n \rangle \tag{28}$$

$$= -\frac{4}{V} \sum_n \langle \tilde{\psi}_n^\alpha | \Delta \psi_n \rangle \tag{29}$$

The quantity inside the square brakets, denoted as $\langle \tilde{\psi}_n^\alpha |$, takes the form of a linear response of $\langle \psi_n |$ under the perturbing potential $[H_{SCF}, r_\alpha]$. Therefore, it can be expressed in the Sternheimer form as shown in Eq. (17).

$$(H_{SCF} - \epsilon_n) \, | \tilde{\psi}_n^\alpha \rangle = -P_c [H_{SCF}, r_\alpha] \, | \psi_n \rangle \, , \tag{30}$$

The perturbing potential for the $|\Delta \psi_n\rangle$ state constists of the electric contribution $V = -e \boldsymbol{\mathcal{E}} \cdot \boldsymbol{r}$ and the Hartree and exchange-correlation contribution

ΔV_{Hxc}, which depends on the perturbed density. The corresponding linear equation is then given by:

$$(H_{SCF} - \epsilon_n) |\Delta\psi_n\rangle = -\sum_\alpha \mathcal{E}_\alpha |\tilde{\psi}_n^\alpha\rangle - \Delta V_{Hxc} |\psi_n\rangle . \tag{31}$$

These equations must be solved self-consistently by updating the screened field $\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{E}}_0 - 4\pi\boldsymbol{\mathcal{P}}$. At each iteration, the polarization in Eq. (29) is calculated, and the screened field is used to find the perturbed density in Eq. (13) and wavefunctions in Eq. (31), until self-consistency is achieved. The electronic susceptibility $\chi^\infty_{\alpha\beta}$ can then be obtained direclty as:

$$\chi^\infty_{\alpha\beta} = \frac{\partial \mathcal{P}_\alpha}{\partial \mathcal{E}_\beta} = -\frac{4}{V}\sum_n \langle \tilde{\psi}_n^\alpha | \frac{\partial \psi_n}{\partial \mathcal{E}_\beta} \rangle . \tag{32}$$

Born effective charges have both electronic and ionic contributions. The ionic contribution is straightforward, as the polarization created after an ion displacement is $\boldsymbol{\mathcal{P}} = Z_s \boldsymbol{u}_s$ and the derivative is simply the charge of the ion Z_s. The electronic contribution is trivial once the linear responses of the wavefunctions to the density $\partial\psi_n/\partial \boldsymbol{u}_s$ are known. In this case, the electronic polarization in Eq. (25) is calculated with respect to the displacement $u_s^\beta(\boldsymbol{q} = 0)$. By summing up the two terms, we have:

$$Z^*_{s,\alpha\beta} = \frac{\partial \mathcal{P}^{tot}_\alpha}{u_s^\beta(\boldsymbol{q}=0)} = Z_s + \frac{4}{V}\sum_n \langle \tilde{\psi}_n^\alpha | \frac{\partial \psi_n}{\partial u_s^\beta(\boldsymbol{q}=0)} \rangle . \tag{33}$$

These two responses are the only ingredients needed to determine the correct frequency of LO phonons in the limit $\boldsymbol{q} \to 0$ in polar insulators. The restoring force acting on ions, once the phonon frequency is known, is of the form $\boldsymbol{F} = -M\omega^2\boldsymbol{u}$. LO phonons are capable of generating equally oriented dipoles, which give rise to an electric field and an additional electrostatic term $Z^*\boldsymbol{\mathcal{E}}$ to the force. The third Maxwell's equation $\boldsymbol{q} \times \boldsymbol{\mathcal{E}} = 0$, ensures that this field is absent for transverse modes, where the only valid solution is $\boldsymbol{\mathcal{E}} = 0$. In the absence of free charges, the divergence of the polarization field $\boldsymbol{\mathcal{D}} = \boldsymbol{\mathcal{E}} + 4\pi\boldsymbol{\mathcal{P}}$ is zero. In highly symmetric systems, $\boldsymbol{q} \cdot \boldsymbol{\mathcal{D}} = 0$ gives $\boldsymbol{\mathcal{D}} = 0$, although this result can be generalized to all crystals [7]. Expanding the polarization

$$\boldsymbol{\mathcal{P}} = \sum_s \frac{\partial \boldsymbol{\mathcal{P}}}{\partial \boldsymbol{u}_s}\boldsymbol{u_s} + \frac{\partial \boldsymbol{\mathcal{P}}}{\partial \boldsymbol{\mathcal{E}}}\boldsymbol{\mathcal{E}} = \sum_s Z_s^* \boldsymbol{u}_s + \chi^\infty \boldsymbol{\mathcal{E}} \tag{34}$$

and defining the electronic contribution of the dielectric function as $\epsilon^\infty = 1 + 4\pi\chi^\infty$ we can determine the electric field as:

$$\boldsymbol{\mathcal{E}} = -\epsilon_\infty^{-1} \sum_s Z_s^* \boldsymbol{u}_s \tag{35}$$

which is responsible for the increased stiffness in longitudinal oscillations. This is why LO phonons have higher frequency in the long-wavelength limit. IFCs also are available from the reciprocal version of Eq. (12) once $\partial\psi_n/\partial\boldsymbol{u}_s(\boldsymbol{q})$ are known. After IFCs are computed for a coarse grid of $\boldsymbol{q}$ points, one can verify the convergence by examining the IFCs in real space using Fast Fourier Transform (FFT). In the real space, the IFCs on a grid of N $\boldsymbol{q}$ points correspond to the force constants in a supercell composed of N unit cells. Tipically, the force constants decay exponentially. Assuming that IFCs in real space are exactly zero outside the supercell, one can obtain by inverse FFT a smooth phonon dispersion by performing an inverse FFT, a procedure known as *Fourier interpolation*. In polar insulators near the Γ point, as mentioned earlier, the dipoles give rise to a long-range polynomial interaction, which introduces an additional non-analytic term in the reciprocal IFCs. The divergence can resolved by calculating the non-analytical part of the force constants from ϵ^{∞} and Z^* [8]. This term is subtracted from the calculated reciprocal IFC before performing FFT, and then re-added later [9].

2 Simulation of a Raman Spectrum

In this section, we will discuss the key elements necessary for constructing a Raman spectrum using the Density Functional Perturbation Theory (DFPT) formalism. Raman scattering phenomena are directly related to a specific subset of the phonon spectrum, which is determined by analyzing the symmetries of the displacement patterns. Peaks in the Raman spectrum occur at energies corresponding to the active phonons, and their intensity is proportional to the matrix element of the perturbation. The natural width of the Raman peaks is inversely proportional to the lifetimes of the phonons. Phonon lifetimes can be affected by various mechanisms that perturb the harmonic Hamiltonian, leading to the degradation of the phonon's stationary states. The calculation of these quantities, such as phonon frequencies, intensities, and lifetimes, can be accomplished within the framework of DFPT.

2.1 Raman scattering

Raman scattering in a crystal is the inelastic scattering of light by lattice vibrations. Once a sample is illuminated with monochromatic light, a small number of photons are adsorbed and re-emitted with a shift towards lower or higher frequencies, giving rise to Stokes and anti-Stokes lines, respectively, in the outgoing spectrum. At the fundamental level, a photon is capable of

exciting or absorbing a normal vibrational mode with a specific cross-section, which is determined by the Raman tensor

$$R_{\alpha\beta}^{\gamma k} = \frac{\partial^3 E}{\partial \mathcal{E}_\alpha \partial \mathcal{E}_\beta \partial u^{\gamma k}} = \frac{\partial \chi_{\alpha\beta}}{\partial u^{\gamma k}}, \tag{36}$$

where greek letters express cartesian directions and k refers to the specific ion of the unit cell. From a phenomenological point of view, the crystal respond to an oscillating field $\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{E}}_0 \cos(\omega t)$ by developing a polarization

$$\boldsymbol{\mathcal{P}} = \chi \boldsymbol{\mathcal{E}} = \left(\chi_0 + \frac{\partial \chi}{\partial \boldsymbol{u}} \cos(\omega_{\boldsymbol{q}} t) \right) \boldsymbol{\mathcal{E}}_0 \cos(\omega t) \tag{37}$$

$$= \chi_0 \boldsymbol{\mathcal{E}}_0 \cos(\omega t) + \frac{1}{2} \frac{\partial \chi}{\partial \boldsymbol{u}} \boldsymbol{\mathcal{E}}_0 [\cos(\omega + \omega_{\boldsymbol{q}}) t + \cos(\omega - \omega_{\boldsymbol{q}}) t]. \tag{38}$$

The second term in the equation is responsible for the frequency shift of the incident light. The photon frequency is chosen to be higher than the frequency of the most energetic phonon, typically in the infrared region. Even at slightly higher frequencies, the incident light has a wavelength much larger than the unit cell length of the crystal. In this limit, we are only probing the region of the phonon spectrum near the Γ point, where the phonon wavevector $\boldsymbol{q} = \boldsymbol{k}_i - \boldsymbol{k}_f$ can be safely assumed to be zero in order to guarantee the conservation of momentum. One can consider the next leading term in the expansion of the susceptibility, which gives rise to a contribution called the non-linear Raman response. However, these higher-order perturbations will be neglected throughout this entire dissertation. Certain modes are characterized by a quadratic susceptibility, where $\partial \chi / \partial \boldsymbol{u} = 0$, such as the out-of-phase (asymmetric) linear stretch of the CO_2 molecule. In this case, opposing dipoles arise as a consequence of the motion of the carbon atom towards the two oxygen atoms. Thus, $\partial \boldsymbol{\mathcal{P}} / \partial \boldsymbol{u} \neq 0$, but the polarization is maximum at $\boldsymbol{u} = 0$ and decreases symmetrically as the carbon atom approaches the two different oxygen atoms. The symmetry of the system then imposes that the derivative of the polarizability of the molecule calculated at zero displacement vanishes, resulting in a severely damped intensity of the Raman line.

2.2 Calculation of Raman intensities

The interaction between the electric field and the electronic cloud is complex. In principle, the photon is capable of exciting the crystal into a virtual state, which is described in terms of the crystal's excited states. However, a simpler approach, known as the *Placzek approximation* [10], assumes that

the electrons are in their respective ground state in both the initial and final states. The ions do not directly respond to the external electric field due to its rapid oscillation. The coupling between phonons and light occurs in two steps: first, the electrons couple with light, and their response is characterized by the electronic susceptibility χ^{∞}; then, the change induced in the electronic density couples with nuclear motion. In this approach, the susceptibility is calculated for each ionic configuration while keeping the ions fixed in their positions. The derivative of the susceptibility can be obtained directly using finite difference calculations. This mechanism follows from the adiabatic approximation: the scattered light intensity during nuclear vibration is assumed to be the same as that obtained by keeping the ion positions fixed at each step of the vibration.

Eq. (38) shows that the dipole oscillates at three different frequencies. Classically, an oscillating dipole generates an oscillating electromagnetic field with a characteristic intensity, polarization, and angular dependence. However, the interaction responsible for the generation of the dipole is treated quantum-mechanically. According to the Fermi golden rule, the intensity is proportional to the transition rate, which, in turn, is proportional to the square of the dipole transition matrix. The matrix involves a specific vibrational mode of branch ν, so the indices γk in Eq. (36) must be summed over the corresponding phonon eigenvector $u_{\nu}^{\gamma k}$ to obtain $R_{\alpha\beta}^{\nu}$. The perturbing Hamiltonian is also expected to include two photon operators and one phonon creation/destruction operator $a_{\nu}^{\dagger}$ and a_{ν}, respectively, for the Stokes and anti-Stokes lines. These operators act between the states $|n_{\nu}\rangle$ and $|n_{\nu} \pm 1\rangle$, contributing as

$$\langle n_{\nu} - 1|a_{\nu}|n_{\nu}\rangle \sim \omega_{\nu}^{-1/2} n_{\nu}^{1/2} \tag{39}$$

$$\langle n_{\nu} + 1|a_{\nu}^{\dagger}|n_{\nu}\rangle \sim \omega_{\nu}^{-1/2} (n_{\nu} + 1)^{1/2}, \tag{40}$$

where n_{ν} is the Bose-Einstein distribution at a given temperature. The resulting light intensity, after having saturated the two indexes of the field with the polarization directions $\boldsymbol{e}$ specific for the experimental setup, is then

$$\frac{I}{I_0} = C \left| \boldsymbol{e}_{\alpha} \cdot R_{\alpha\beta}^{\nu} \cdot \boldsymbol{e}_{\beta} \right|^2 \frac{n_{\nu}\,(+1)}{\omega_{\nu}}. \tag{41}$$

I_0 is the incident light intensity and C is a constant that reflects the behaviour of the oscillating dipole. Classically, $C \sim (\omega \pm \omega_{\nu})^4$, thus this term is constant only if $\omega \gg \omega_{\nu}$, otherwise this correction must be taken into account.

The difficulty in the calculation of Raman intensities is hindered by the mixed third derivative of the energy in Eq. (36). A well-known theorem, valid within

perturbation theory, is the *2n+1 theorem* [11]: according to it, the $2n+1$ derivative of the energy can be constructed from the n-th derivative of the eigenvectors. This theorem can be adapted in the DFTP framework without introducing additional computational complexity [12]. In our case, after calculating the phonons and dielectric constants, both $\partial\psi/\partial\boldsymbol{u}$ and $\partial\psi/\partial\boldsymbol{\mathcal{E}}$ are available. Let's first analyze the energy derivative with respect to two fields of direction α and β, denoted as $E^{\alpha\beta}$. Both the Hartree and exchange-correlation terms do not explicitly depend on the electric field. Moreover, the perturbation $-e\boldsymbol{r}\cdot\boldsymbol{\mathcal{E}}$ is linear in the field, resulting in the second-order correction vanishing. In a Taylor series, the second derivative is twice the coefficient of the expansion. We decompose the derivative into two pieces as $E^{\alpha\beta} = 2(E^{\alpha\beta}_{\text{lin}} + E^{\alpha\beta}_{\text{quad}})$, where [12]:

$$E^{\alpha\beta}_{\text{lin}} = -\sum_v \langle\psi_v|r_\alpha|\psi^\beta_v\rangle + \text{c. c.} = V\chi_{\alpha\beta} \tag{42}$$

$$E^{\alpha\beta}_{\text{quad}} = \sum_v \langle\psi^\alpha_v|H-\epsilon_v|\psi^\beta_v\rangle + \frac{1}{2}\int\int d^3r d^3r' \frac{\delta^2 E_{Hxc}[n]}{\delta n(\boldsymbol{r})\delta n(\boldsymbol{r'})} n^\alpha(\boldsymbol{r}) n^\beta(\boldsymbol{r'}). \tag{43}$$

The superscript is intended to represent a derivative, where the linear and quadratic contributions are intended with respect to the derivative of the wavefunctions. Eq. (32) has been used to expand the linear term. The quadratic term acts as a local field ΔV_{Hxc} in Eq. (31), its effect is calculated self-consistently to obtain the susceptibility. However, the susceptibility itself is exclusively given by the linear term. The self-consistent cycle approximately solves for a correct energy functional. An equivalent way to state the problem is to require the wavefunctions to exactly solve the energy functional, truncated at second order, along with the orthonormality condition as Lagrange multipliers. As a functional of the wavefunction derivatives, the energy is

$$E[\psi^{(1)}] = E^{(0)} + \sum_\alpha E^\alpha[\psi^{(0)}]\mathcal{E}^\alpha + \frac{1}{2}\sum_{\alpha\beta} E^{\alpha\beta}[\psi^{(0)},\psi^{(1)}]\mathcal{E}_\alpha\mathcal{E}_\beta. \tag{44}$$

In functions involving linear and quadratic degrees of freedom, the quadratic term is related to the linear term, once both are calculated at the minimum: $f_{\text{lin}} = -2f_{\text{quad}}$. This property is evident in the one-dimensional case: if $f(x) = ax^2 + bx$, in the minimum $x_0 = -b/2a$ the property holds. When the self-consistent solutions are substituted into both terms, the same relation holds: $E^{\alpha\beta}_{\text{lin}} = -2E^{\alpha\beta}_{\text{quad}}$. This explains why the susceptibility, defined in Eq. (32) and Eq. (42), corresponds to the second-order derivative of the energy: $E^{\alpha\beta} = 2(E^{\alpha\beta}_{\text{lin}} + E^{\alpha\beta}_{\text{quad}}) = E^{\alpha\beta}_{\text{lin}}$ at the variational minimum. Ultimately, this justifies the equality displayed in Eq. (36).

The Raman tensor can be calculated by finite differences through performing $36N_{\text{at}}$ susceptibility calculations, where N_{at} is the number of atoms in the unit cell. DFPT divides into three equivalent approaches:

- In the first approach, the mixed third derivative of the energy is calculated using the 2n+1 formula. The resulting coefficient $R^{\alpha\beta\gamma}$, where γ is the displacement coordinate, is the sum of the six permutations of the form $E^{\alpha\beta\gamma}$. In Ref. [13], the authors evaluate the expressions for the six 2n+1 terms. They use the Berry phase formalism, in which the position operator is replaced with $-i\partial/\partial\boldsymbol{k}$ acting on the periodic part of the Bloch functions. A clever discretization technique must be introduced to evaluate the derivative numerically, as well as Bloch functions over a dense grid of $\boldsymbol{k}$-points.

- In the second appproach, the Raman tensor is evaluated direclty using to $\psi^{\alpha\beta}$ [14] as follows:

$$\frac{\partial^3 E}{\partial\mathcal{E}_\alpha\partial\mathcal{E}_\beta\partial u^{k\gamma}} = \int d^3r\, \frac{\partial V_{ext}(\boldsymbol{r})}{\partial u^{k\gamma}} \frac{\partial^2 n}{\partial\mathcal{E}_\alpha\mathcal{E}_\beta}(\boldsymbol{r}), \tag{45}$$

  which is obtained by differentiating the force in Eq. (11) with respect to the two fields. The external potential is the ionic one, hence it does not depend explicitly on the electric fields. The phonon derivative $\psi^{k\gamma}$ is not required. This approach involves calculating six second order wavefunctions through a Sternheimer equation. The entire problem can be reformulated in terms of the density matrix [15], equal to the projection operator onto the valence band. The density matrix has two advantages: (i) gauge freedom in the phase of KS orbitals is removed, resulting in naturally gauge-invariant equations; (ii) the off-diagonal matrix elements in real space decay exponentially, so the density is localized and the action of position operator well defined.

- In the third approach, the Raman tensor is directly evaluated as in the previous approach and $\psi^{\alpha\beta}$ is obtained from the usual Sternheimer equation acting on KS orbitals. The usual gauge choice is the so-called *parrallel transport gauge*, according to which the higher derivative projections onto the valence band are minimized: $\langle\psi_v^{(1)}|\psi_w^{(0)}\rangle = 0$ and $\langle\psi_v^{(2)}|\psi_w^{(0)}\rangle = -\frac{1}{2}\langle\psi_v^{(1)}|\psi_w^{(1)}\rangle$, as a consequence of orthogonality conditions and the gauge definition,

$$\langle\psi_v^{(n)}|\psi_w^{(0)}\rangle - \langle\psi_v^{(0)}|\psi_w^{(n)}\rangle = 0 \tag{46}$$

for each v and w in the valence band. The relative Sternheimer equation is [12]:

$$P_c(H^{(0)} - \epsilon_v^{(0)})P_c\,|\psi_v^{(2)}\rangle = -P_c\left(H^{(1)}\,|\psi_v^{(1)}\rangle - \sum_w \Lambda_{vw}^{(1)}\,|\psi_w^{(1)}\rangle\right). \quad (47)$$

Λ is the Lagrange multiplier that guarantees orthogonality, $\Lambda_{vw}^{(1)} = \langle\psi_v^{(0)}|H^{(1)}|\psi_w^{(0)}\rangle$; the second-order terms disappear from the right side after the projection. In electric field calculations, the term $\Lambda_{vv}^{(1)} = -e\,\langle\psi_v^{(0)}|r_\alpha|\psi_v^{(0)}\rangle$ is undefined, so the projection $\langle\psi_v^{(1)}|\psi_v^{(2)}\rangle$ will be unknown.

We used a code based on the second approach. Raman spectroscopy is usually conducted on powdered samples, the intensity should involve averaging over all the possible orientations of the crystal. Both the integrated and the differential intensities over a solid angle are functions of rotational invariants [16] [17], which are specific combinations of the Raman tensor. In some low-symmetry polar insulators, the frequency of LO phonons depends on the direction of the $\boldsymbol{q}$ vector near the Γ point. $\boldsymbol{q} = \boldsymbol{k}_i - \boldsymbol{k}_f$ is not exactly zero and takes all the directions during the averaging process.

2.3 Natural width of Raman lines

In the harmonic approximation, perfect crystals lack a scattering mechanism for phonons. Without this mechanism, phonons exist as stationary states of the Hamiltonian and possess infinite lifetimes. Consequently, phonons cannot relax to the Bose-Einstein distribution, and the concept of temperature remains undefined. The two most significant scattering mechanisms are scattering due to lattice defects and anharmonic scattering. Both can be treated perturbatively. The first perturbing potential arises from the mixing of the lattice periodic potential with the substitution potential, where the expansion parameter is the substitution percentage. The second mechanism is described by higher-order terms in the Taylor expansion of the potential around the equilibrium point. Each order beyond the second can be treated perturbatively using green function formalism, with the harmonic phonons as a basis. For each phonon one can obtain a self-energy comprising a real and an imaginary part. The real part affects the frequency of the phonon, the imaginary part introduces a lifetime of the state. Typically, only the imaginary part of the third order is considered, since the shifts caused by the real part of the third and fourth order tend to cancel each other out. In our specific situation, Raman scattering drives the phonon populations near

Γ out of equilibrium. In the relaxation time approximation, the scattering rate is calculated as follows:

$$\frac{\partial_{\boldsymbol{q}\nu}}{\partial t} \simeq \frac{\tilde{n}_{\boldsymbol{q}\nu} - n_{\boldsymbol{q}\nu}}{\tau_{\boldsymbol{q}\nu}}, \tag{48}$$

where $\tilde{n}_{\boldsymbol{q}\nu}$ represents the non-equilibrium number of phonons with momentum $\boldsymbol{q}$ in the branch ν, $n_{\boldsymbol{q}\nu}$ denotes the Bose-Einstein equilibrium distribution, and $\tau_{\boldsymbol{q}\nu}$ corresponds to the relaxation time. This approximation assumes that the distribution of incoming particles within a certain time interval depends solely on equilibrium quantities. In other words, scattering events erase all information about the non-equilibrium state. The scattering rate is determined using the Fermi's golden rule, which states:

$$P_i^f(\text{3ph}) = 2\pi |\langle i|V_3|f\rangle|^2 \delta(E_f - E_i). \tag{49}$$

Both the 3-phonon processes responsible for the creation and annihilation of a phonon in the branch of interest must be considered. The matrix element is calculated in the number representation once the third-order anharmonic term is expressed in terms of creation and destruction operators. Phonons can be expressed as displacements in real space $\boldsymbol{u}_s(\boldsymbol{l})$ or in reciprocal space as $\boldsymbol{u}_s(\boldsymbol{q})$. The diagonalization of the IFC

$$m_s \omega_{\boldsymbol{q}\nu}^2 \boldsymbol{e}_s(\boldsymbol{q}\nu) = \sum_{s'} C_{ss'}(\boldsymbol{q}) \boldsymbol{e}_{s'}(\boldsymbol{q}\nu) \tag{50}$$

gives ν eigenvalues and eigenvectors, which can be used to express the global phonon displacement as

$$u_{\boldsymbol{q}\nu} = \sum_s \sqrt{M_s} \boldsymbol{e}_s^*(\boldsymbol{q}\nu) \boldsymbol{u}_s(\boldsymbol{q}) = -i \frac{1}{\sqrt{2\omega_{\boldsymbol{q}\nu}}} (a_{\boldsymbol{q}\nu}^\dagger - a_{-\boldsymbol{q}\nu}). \tag{51}$$

The anharmonic matrix elemenet is then

$$V_3 = \frac{1}{3!} \sum_{\boldsymbol{q}\nu, \boldsymbol{q'}\nu', \boldsymbol{q''}\nu''} \delta_{\boldsymbol{G}, \boldsymbol{q}+\boldsymbol{q'}+\boldsymbol{q''}} \tilde{V}_3(\boldsymbol{q}\nu, \boldsymbol{q'}\nu', \boldsymbol{q''}\nu'') \tag{52}$$

$$\times (a_{\boldsymbol{q}\nu}^\dagger - a_{-\boldsymbol{q}\nu})(a_{\boldsymbol{q'}\nu'}^\dagger - a_{-\boldsymbol{q'}\nu'})(a_{\boldsymbol{q''}\nu''}^\dagger - a_{-\boldsymbol{q''}\nu''}), \tag{53}$$

where $\tilde{V}_3$ is transformed from the 3-body IFC $C_{ss's''}(\boldsymbol{q}, \boldsymbol{q'}, \boldsymbol{q''})$ as in Eq. (51); the delta is added to satisfy momentum conservation, $\boldsymbol{G}$ is a reciprocal lattice vector. Eq. (53) allows for six combinations of 3-phonon scattering, divided into three classes, since each combination has its conjugate that cancels it at equilibrium. Moreover, the term under summation is invariant under a

permutation of the set $(\boldsymbol{q}\nu, \boldsymbol{q'}\nu', \boldsymbol{q''}\nu'')$: the derivative $\tilde{V_3}$ is cyclic, and the operators commute if the elements in the set are different. In our specific situation, we are studying the scattering from and towards Γ. Therefore, we fix the set as $(\mathbf{0}\nu, \boldsymbol{q'}\nu', \boldsymbol{q''}\nu'')$ and remove the $1/6$ prefactor since there are six permutations. Umklapp scattering $\boldsymbol{G} = \boldsymbol{q'} + \boldsymbol{q''}$ occurs only at the Brillouin zone boundaries. It can be neglected by setting $\boldsymbol{G} = 0$ and $\boldsymbol{q''} = -\boldsymbol{q'}$. The anharmonic term becomes

$$V_3 = \sum_{\nu\nu'\nu'',\boldsymbol{q}} \tilde{V_3}(\mathbf{0}\nu, \boldsymbol{q}\nu', -\boldsymbol{q}\nu'')(a^\dagger_{\mathbf{0}\nu} - a_{\mathbf{0}\nu})(a^\dagger_{\boldsymbol{q}\nu'} - a_{-\boldsymbol{q}\nu'})(a^\dagger_{-\boldsymbol{q}\nu''} - a_{\boldsymbol{q}\nu''}). \quad (54)$$

The initial state after Raman scattering, denoted as $|i\rangle$, has all populations at thermal equilibrium, except for those at Γ. We can further simplify the scenario by neglecting the process $\mathbf{0} + \mathbf{0} \to \mathbf{0}$, reducing it to a particular case of the Single Mode Relaxation Time (SMRT) approximation. In this approximation, we assume that for each mode, all the other modes participating in the scattering process are at thermal equilibrium. Consequently, the initial state can be expressed as $|n_{\mathbf{0}\nu} + \delta_\nu, ...\rangle$, where δ_ν represents a small displacement from equilibrium, and all the other modes (including those at Γ) are set to equilibrium. For class 1 scattering elements, we have the following expressions:

$$P^1_{\boldsymbol{q}\nu\nu'\nu''} = 2\pi|\tilde{V_3}(\mathbf{0}\nu, \boldsymbol{q}\nu', -\boldsymbol{q}\nu'')|^2[|\langle a^\dagger_{\mathbf{0}\nu} a^\dagger_{\boldsymbol{q}\nu'} a_{\boldsymbol{q}\nu''}\rangle|^2 \delta(\omega_{\mathbf{0}\nu} + \omega_{\boldsymbol{q}\nu'} - \omega_{\boldsymbol{q}\nu''}) \quad (55)$$

$$-|\langle a_{\mathbf{0}\nu} a_{-\boldsymbol{q}\nu'} a^\dagger_{-\boldsymbol{q}\nu''}\rangle|^2 \delta(\omega_{\mathbf{0}\nu} + \omega_{-\boldsymbol{q}\nu'} - \omega_{-\boldsymbol{q}\nu''})] \quad (56)$$

From the definition of the Fourier transform of the three-body IFC we also have the property

$$|\tilde{V_3}(\mathbf{0}\nu, \boldsymbol{q}\nu', -\boldsymbol{q}\nu'')|^2 = |\tilde{V_3}(\mathbf{0}\nu, -\boldsymbol{q}\nu', \boldsymbol{q}\nu'')|^2, \quad (57)$$

which allows us to couple each first terms in Eq. (55) with second terms in Eq. (56) at $\boldsymbol{q} \to -\boldsymbol{q}$:

$$P^1_{\boldsymbol{q}\nu\nu'\nu''} = 2\pi|\tilde{V_3}(\mathbf{0}\nu, \boldsymbol{q}\nu', -\boldsymbol{q}\nu'')|^2 \delta(\omega_{\mathbf{0}\nu} + \omega_{\boldsymbol{q}\nu'} - \omega_{\boldsymbol{q}\nu''}) \quad (58)$$

$$[(n_{\mathbf{0}\nu} + \delta_\nu + 1)(n_{\boldsymbol{q}\nu'} + 1)n_{\boldsymbol{q}\nu''} - (n_{\mathbf{0}\nu} + \delta_\nu)n_{\boldsymbol{q}\nu'}(n_{\boldsymbol{q}\nu''} + 1)]. \quad (59)$$

Since at equilibrium each process is balanced with its conjugate, terms in Eq. (59) which does not contain δ_ν should disappear, giving $[\delta_\nu(n_{\boldsymbol{q}\nu''} - n_{\boldsymbol{q}\nu'})]$. Class 2 processes are obtained by switching ν' with ν'': they give the same contribution after $\boldsymbol{q}\nu'\nu''$ are summed up. Class 3 processes are of the type

$a^\dagger_{\mathbf{0}\nu} a_{-\boldsymbol{q}\nu'} a_{\boldsymbol{q}\nu''}$, they contribute as $[-\delta_\nu(n_{\boldsymbol{q}\nu'} + n_{-\boldsymbol{q}\nu''} + 1)]$ with energy conservation $\omega_{\mathbf{0}\nu} - \omega_{\boldsymbol{q}\nu'} - \omega_{-\boldsymbol{q}\nu''} = 0$. We can finally express the formula of the linewidth, summing the contributions of the three classes:

$$\Gamma_\nu = \frac{1}{\tau_\nu} = \frac{1}{\tilde{n}_{\mathbf{0}\nu} - n_{\mathbf{0}\nu}} \frac{\partial \tilde{n}_{\mathbf{0}\nu}}{\partial t} = \frac{P_\nu}{\delta_\nu} \tag{60}$$

$$= 2\pi \sum_{\boldsymbol{q}\nu'\nu''} |\tilde{V}_3(\mathbf{0}\nu, \boldsymbol{q}\nu', -\boldsymbol{q}\nu'')|^2 [2(n_{\boldsymbol{q}\nu''} - n_{\boldsymbol{q}\nu'})\delta(\omega_{\mathbf{0}\nu} + \omega_{\boldsymbol{q}\nu'} - \omega_{\boldsymbol{q}\nu''}) \tag{61}$$

$$-(n_{\boldsymbol{q}\nu'} + n_{-\boldsymbol{q}\nu''} + 1)\delta(\omega_{\mathbf{0}\nu} - \omega_{\boldsymbol{q}\nu'} - \omega_{-\boldsymbol{q}\nu''})]. \tag{62}$$

To perform the summation, a dense grid of $\boldsymbol{q}$ points is required. Three-body IFCs are calculated on small grids, and then interpolated using similar methods as in the two-body case. Delta functions are substituted with gaussians of arbitrary width.
The time-consuming operation relies on the DFPT calculation of the third order energy derivatives. These quantities can be directly accessed from 2n+1 theorem, but the following complications arise:

- The perturbing potential $\Delta V^{\boldsymbol{q}}$ carries a momentum $\boldsymbol{q}$, then the third order terms of the kind $\langle \psi^{(1)} | H^{(1)} | \psi^{(1)} \rangle$ [18] require the knowledge of the $\boldsymbol{q}_1$ perturbed wavefunctions at $\boldsymbol{k} - \boldsymbol{q}_2$ k-points, where $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$ assume all the possible combinations $(\boldsymbol{q}, \boldsymbol{q'}, \boldsymbol{q''})$ in the calculation of $E^{\boldsymbol{q},\boldsymbol{q'},\boldsymbol{q''}}$. The calculation involves solving non-self-consistent Sternheimer equations after finding the self-consistent perturbing potential. However, this procedure requires significantly more CPU time compared to the self-consistent counterpart.

- In metals, smearing is introduced to achieve faster convergence with respect to the k-points grid in the KS ground state calculation. However, this technique introduces spurious terms in the DFPT energy expansion [19], and the number of these terms increases when considering higher-order derivatives.

- The third derivative of exchange-correlation functionals with respect to density contain a huge amount of terms in gradient-corrected functionals.

- Derivatives of the external potential include additional contributions for non-local pseudopotentials. Non-linear core corrections also contribute with additional terms originating from the core charge and its derivatives.

These complications add complexity and computational cost to the calculation of third-order energy derivatives in DFPT.

3 Raman spectrum of Corundum

3.1 Computational methods and Results

All the computations are performed with the DFT Quantum ESPRESSO package [20, 21, 22], using PBEsol GGA functional [23] and norm-conserving pseudopotentials [24, 25]. Electronic wavefunctions and charge densities are expanded in plane waves with a kinetic energy cutoff of 90 Ry and 360 Ry cutoff, respectively. All structures are relaxed until the threshold of 10^{-4} Ry for the total energy and 10^{-4} Ry/a_0 for each component of the force acting on the ions is reached. The cell is free to change during the relaxation, with the constraint to remain in the rhombohedral lattice system. Born effective charges and dielectric constant are obtained from the DFPT mechanism discussed in the previous sessions. For corundrum, a shifted $4\times4\times4$ k-points grid is used, calculations are performed in the unit cell. Both the cutoff and the number of k-points are converged with a threshold of 10^{-4} Ry. Phonon calculations are performed in a $4 \times 4 \times 4$ q-points grid. This grid is useful to plot the phonon's dispersion and to calculate the anharmonicity term, while Raman frequencies and cross sections are calculated just with phonons at Γ point. LO-TO splitting is calculated in the (1,1,1) direction, as in [8]. The resulting spectrum is displayed in Fig. (1).

4 Argon impurities in Corundum

4.1 Defect types

This study is inserted in a broader investigation of various deep minerals which are thought being effective in trapping noble gases. In this context, various configurations of Argon defects inside Stishovite (SiO_2) and Corundum (Al_2O_3) are studied. These systems are synthesized in chambers, adding gaseous Ar and increasing pressure. Composition and aspect of the resulting materials are analysed at atmospheric pressures through scanning electron microscopy (SEM), then both IR and Raman spectra are acquired. In Stishovites, the addition of Argon forces the structure to decompose into a brittle phase, in which Ar is found to be gaseous and completely absent inside the material. DFT calculations for Stishovite are performed for the following point defects: Ar_{Si}, Ar_O substitutions and Ar_i interstitial. Neither of these structures results physically suited for a Raman calculation, due to the appearence of a metallic band inside the gap. Corundum, on the other hand, remains stable after the addition of Ar, SEM proves that Ar is present everywhere in the sample with high homogeneity. X-Ray Diffraction analysis

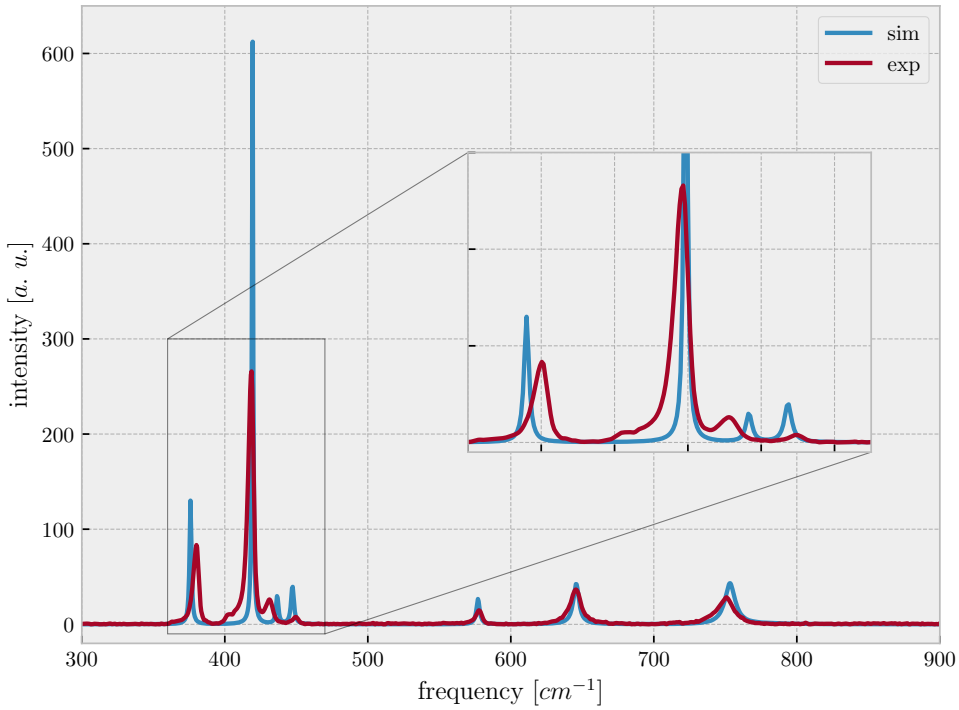


Figure 1: Raman spectrum of Corundum (Al_2O_3). Experimental spectrum in red is taken from Ref. [26], the blue line is simulated with linewidths obtained from anharmonicity. Frequencies of the simulated spectrum are rescaled by 0.8 %.

shows that the structure is crystalline and the Ar is found in the interstitial position. For these reason, we will focus only on Corundum from now on.

4.2 Defect characterization

DFT calculations for Corundum include the following defects: Ar_{Al}, Ar_O substitutions and Ar_i interstitial. Defects are studied in a $2 \times 2 \times 2$ supercell, to avoid interactions between different copies; k-points grid is rescaled to keep constant the density of k-points. All the relaxations are (collinearly) spin polarized, charged substitutions up to the oxidation number of the species are calculated as well. Because of its closed shell configuration, Ar is a noble gas, i.e. it is very unlikely to interact strongly and form chemical bonds with other species, even when put in crystalline medium. However, at the local-density DFT level the functional form of the exchange is not exact, being the exact exchange a non-local operator. This error introduces self interactions which favor fractional occupations. Ar pseudopotential has valence $3s^2 3p^6$,

	direct (eV)	vacancy (eV)	$\Delta V/V$
Ar_i	11.4	11.4	2.9%
Ar_{Al}	16.0	3.6	1.8%
Ar_O	17.9	10.9	3.8%

Table 1: Formation energies of the defects. Direct approach means calculating chemical potentials as energies of the most stable phase of the element, vacancy approach means calculating the energy of the vacancy instead of the chemical potential of the element. The last column refers to the relative volume difference

but other combinations can be tested. The most naïve approach to avoid the self-interaction problem is to freeze localized electrons in the pseudopotential core. We tried to take this method to its more extreme by using a Ar pseudopotentials with zero electrons, but it proved to yield unpredictive results.
The formation energy of the interstitial is given directly by

$$\Delta E(\mathrm{Ar_i}) = E(\mathrm{Ar_i}) - \mu(\mathrm{Ar}) - E(\mathrm{Al_2O_3}), \tag{63}$$

where μ is the chemical potential of the added Argon, calculated as the energy of a single Argon atom in a cubic supercell with a length of 10 Å. Formation energy of substitutions are generally calculated knowing the chemical potential of all the species that participate in the reaction. For charged defects with charge q, the electronic chemical potential – i. e. the Fermi energy E_f – plays its role [27]:

$$\Delta E(\mathrm{Ar_X^q}) = E(\mathrm{Ar_X^q}) - E(\mathrm{Al_2O_3}) - \mu(\mathrm{Ar}) + \mu(\mathrm{X}) + qE_f, \tag{64}$$

where X = Al, O. Chemical potentials of the single species forming a compound are sensible to the growing conditions [28]: during the growth of Al_2O_3, at the equilibrium $\mu(Al_2O_3) = 2\mu(Al) + 3\mu(O)$, where $\mu(Al_2O_3)$ is the formation energy of the compound. Knowing that compounds with same species and different chemical compositions are not growing, one can put narrow limits on the chemical potential of the single species. An alternative and more direct approach is to calculate the ground state energy of species in their most stable environment: in this case $\mu(O) = 1/2E(O_2^{mol})$ and $\mu(Al) = E(Al^{bulk})$. One last approach is to substitute to $E(Al_2O_3) - \mu(X)$ the ground state energy of the vacancy $E(V_X)$ [29]. Results are summarized in Tab. (1), along with the relative expansion of the cell after the formation of the defect.

The two methods are in total disagreement between each other: the calculated energy of the vacancy results higher even with polarized spins; moreover, formation energies are all exceptionally high. This can be due to the

different thermodynamic conditions: during the synthesis the capsule was kept at 10.5 GPa and 400 K, while our DFT calculations are all performed at $T = 0$ K and zero pressure. This choice has been made to compare with the spectral measurement which are made after the pressure is relieved. We also neglected the zero point energy of phonons, thermodynamic corrections for the gases and entropic contributions, but these should account for less than 1 eV. It's also possible that pre-existing defects facilitate the entrance of Ar. In any case, XRD data confirms that Ar is found on the interstitial position; DFT predicts a metallic state for both substitutions, hence Raman spectra can be computed exclusively for Ar_i. In this context, charged cell calculations were performed only to check if a different oxidation state was able to produce a non-metallic ground state, giving no interesting result for neither of the two substitutions.

4.3 Electronic structure of Al_2O_3 with interstitial Ar

First we start by analysing the electronic structure of the Ar_i defect. The ground state is spin unpolarized and non-metallic, in Fig. (2) it's plotted the corresponding normalized density of states (DOS) and its decomposition as projection over oxygen p-orbitals. As we can see, electrons in the impurity band at ~ -2.5 eV are mostly in Oxygen p-orbitals. The residual DOS is attributed to Argon orbitals, as we can clearly see in Fig. (3). Only Ar p-orbitals projection is non-zero in the band; the resulting spherical charge distribution shows that the three orbitals are equally populated, while Oxygen orbitals are all pointed in the direction of the Argon. The formation of this impurity band shows that the 8 valence electrons of Argon are interacting with the surrounding Oxygen. Ar is found in an octahedral environment, with three couples of O atoms that are collinear with Ar. If we displace a single couple symmetrically, we would expect a change in the polarizability of the system, just like in the in-phase stretch vibrational mode of the CO_2: we expect then some Raman-active modes related to the presence of the interstitial. Knowing that these signs are not found experimentally, this evidence could be considered as a failure in the functional approximation to correctly describe the charge density near the defect. Ar, being a noble gas, should be an almost chemically-inactive species, while at the DFT level we have the opposite situation. Increasing the supercell size has the effect of eliminating the spurious interaction between two different defects, but it's impossible to completely avoid the interaction between Argon and neighbouring Oxygens. Solution of this inadequacy all involve a mechanism capable of localizing Ar valence electrons. We have seen that the addition of spins does not work, since the ground state is still spin unpolarized. Moreover, Ar_i defect must

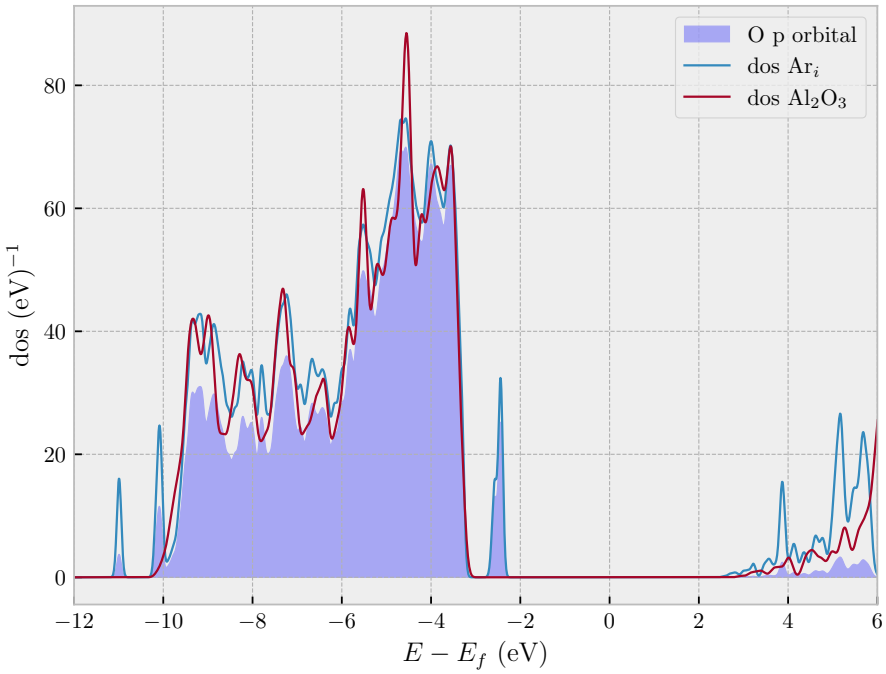


Figure 2: Comparison between interstitial Argon (blue) and crystalline Al_2O_3 (red) density of states. The colored area under interstitial DOS is projected onto Oxygen p-orbitals. The new impurity band across the bandgap is strongly due to Oxygen p-orbitals.

be charge-neutral, since there is no species leaving the material with a different number of electrons. Pseudopotentials with no valence electrons and with fixed s-orbital have been tested, giving always an unstable structure. LDA+U has not been tested, it could be an effective approach to force electrons in Ar p-orbitals to not participate in bindings, as it will encourage the Ar p-electrons to stay localized, by correcting the self-interaction.

4.4 Raman spectra

The number of Raman-active modes in a Raman spectra depends on two factors: the number of atoms in the cell N_{at} and the symmetries of the system. By looking exclusively at the Γ point, the $2 \times 2 \times 2$ has eight times more normal modes. These modes are due to band folding: the first Brillouin Zone is reduced with a factor of eight and the vectors that before were pointing in specific points inside the zone now are reciprocal lattice

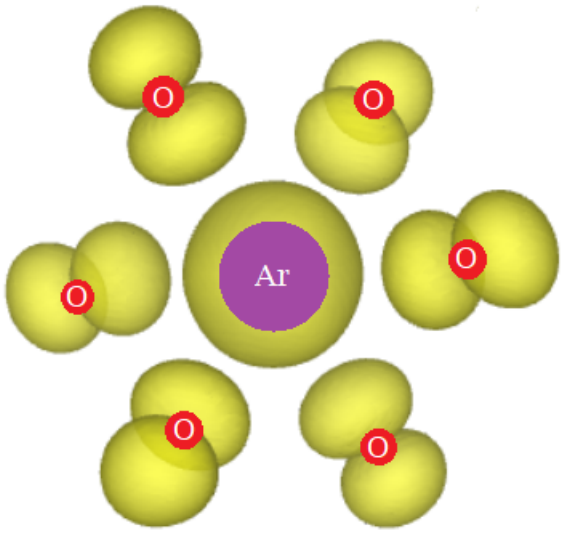


Figure 3: Integrated density of states in the energy interval related to the impurity band. Electrons are concentrated inside the (yellow) iso-density surface. Oxygens form an octahedral environment around Ar ion. The p-orbital character of Oxygens in this band is clearly evident.

vectors. As a consequence, the Γ point of the pristine Al_2O_3 contains phonon frequencies of other seven q-points. After the addition of the interstitial, we are introducing three new modes that mix with modes in each of these q-points; the coupling between modes depends on how much the defect interact with its surroundings.

The symmetry reduction caused by the introduction of a point-defect can activate modes that were not Raman-active in the pristine crystal. However the measured intensity of these modes would not only be proportional to their ab-initio Raman cross section, but should also take into account the relative abundance of defects in the crystal: if only one out of N cells contains an Ar insertion, the intensity of the new modes will be $1/N$ times its computed value. We first performed Raman calculations on a $2 \times 2 \times 2$ supercell: the resulting spectrum, showed in Fig. (4), is characterized by a large amount of lines, all off-centered with respect to the experimental ones. All simulated lines are displayed with a constant width, the anharmonic contribution has not been calculated. One other problem is that all these structures are characterized by phonons with imaginary frequency. Displacing the structure in the direction of one of these modes, the energy lowers: this behaviour is a symptom of mechanical instability. In particular, imaginary phonons at Γ indicate that internal symmetry has to be relaxed, while for out of Γ phonons translational symmetry has to be relaxed, i.e. a larger cell is needed. At

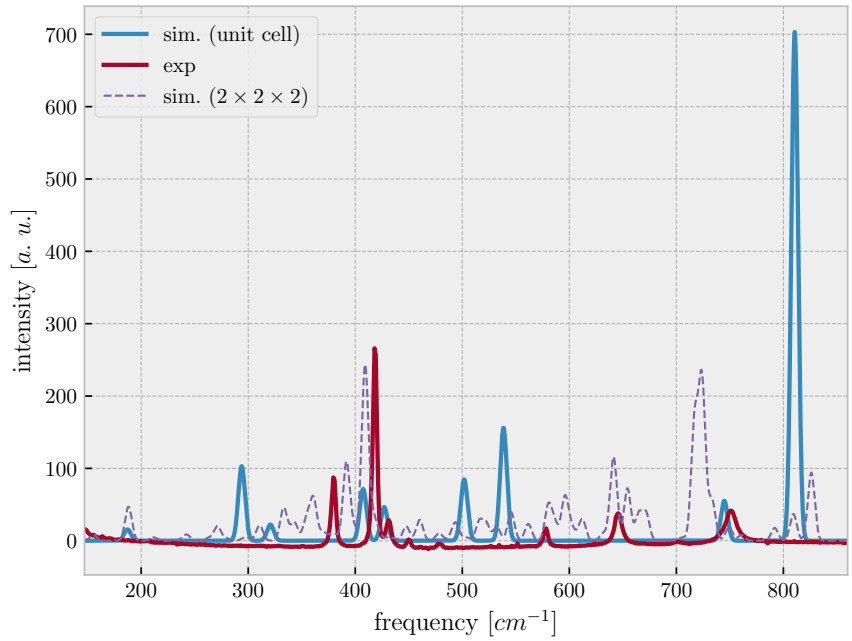


Figure 4: Comparison between experimental (red) and simulated (blue) Raman spectra of the interstitial Ar_i in Al_2O_3. Each simulated line has a constant width, Ar is placed in a unit cell. The $2 \times 2 \times 2$ spectrum (purple dashed) is also present.

the current thermodynamical conditions, it can be avoided by relaxing the structure without any constraint on the cell and then displacing according to the phonon's eigenvector and repeat the procedure until all modes has disappeared. The removal of symmetries results in a higher percentage of Raman-active modes, that in our case are all inconsistent with the experimental result. The other option is to study the defect with its symmetries and in its unit cell, knowing that there will be higher spurious interaction between copies of the site. Even in this case, the results are inconsistent.

In Fig (4) a high peak at $810\,\mathrm{cm}^{-1}$ appear after Ar addition. The relative displacement is showed in Fig. (5b). The most similar displacement in the pristine system, in which the Oxygens move approximately in the direction of the Aluminum, is displayed in Fig. (5a); the first difference between the two systems is that during relaxation the three oxygens move away from the Argon. The second is that the mode becomes active only after the Ar addition, in both cases the displacement corresponds to one of the highest frequency of the system: $680\,\mathrm{cm}^{-1}$ for Al_2O_3 and $810\,\mathrm{cm}^{-1}$ for Ar_i. This ex-

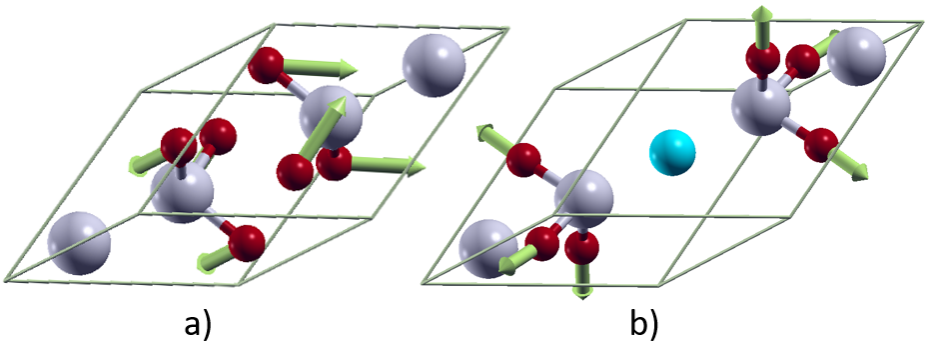


Figure 5: Comparison between two normal modes in a) Al_2O_3 and b) the interstitial Ar_i. The first is at 680 cm^{-1} and it is not Raman-active, while the second has a frequency of 810 cm^{-1} and it's Raman-active.

ample shows that the high structural relaxation changes the symmetry of the modes. Moreover, unit cell and supercell Raman spectra are totally inconsistent: in the second case the relaxation strongly affects only the coordinating Oxygens.

5 Conclusions

We first analyzed the Density Functional Perturbation (DFPT) scheme in extended systems, with particular attention to the calculation of the interatomic force constants (IFCs) to obtain phonon frequencies and eigenvectors. We discussed the intricacies of calculating the system's response to homogeneous electric fields. We then presented some approaches to computing third-order derivatives of the energies, specifically the linear Raman coefficient and the third-order IFCs. Using these components, we simulated the Raman spectrum of Corundum (Al_2O_3), incorporating the phonon linewidths determined by the anharmonic term. In the final section, we focused on simulating Argon impurities in Corundum. We first computed the formation energies of various defects and attempted to construct the Raman spectrum for interstitial Argon. However, the results do not align with the experimental spectrum, likely due to the limited capability of the employed functional approximation (GGA) to accurately capture the binding between Argon and its surroundings.

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